

UE19CS251

DESIGN AND ANALYSIS OF ALGORITHMS

UNIT 5: Limitations of Algorithmic Power and  
Coping with the Limitations

Backtracking

PES University

## Outline

### Concepts covered

- Backtracking
  - Introduction
  - $N$  Queens
  - Hamiltonian Circuit
  - Subset Sum
  - Algorithm

## 1 Tackling Difficult Combinatorial Problems

- There are two principal approaches to tackling difficult combinatorial problems (NP-hard problems):
  - Use a strategy that guarantees solving the problem exactly but doesn't guarantee to find a solution in polynomial time
  - Use an approximation algorithm that can find an approximate (sub-optimal) solution in polynomial time

## 2 Exact Solution Strategies

- Exhaustive search (brute force)

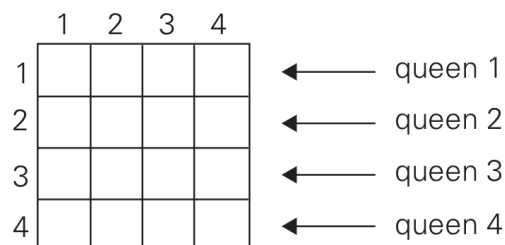
- useful only for small instances
- Dynamic programming
  - applicable to some problems (e.g., the knapsack problem)
- Backtracking
  - eliminates some unnecessary cases from consideration
  - yields solutions in reasonable time for many instances but worst case is still exponential
- Branch-and-bound
  - further refines the backtracking idea for optimization problems

### 3 Introduction

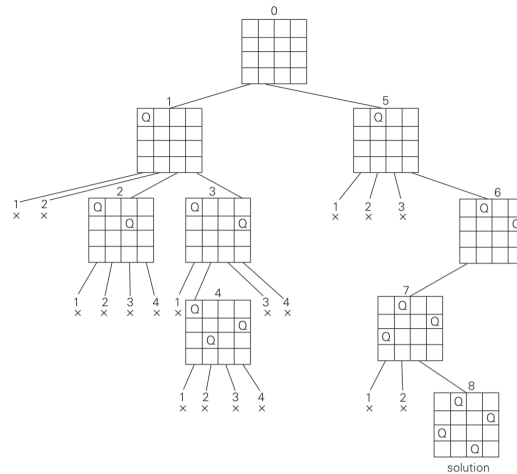
- Construct the *state-space tree*
  - nodes: partial solutions
  - edges: choices in extending partial solutions
- Explore the state space tree using depth-first search
- “Prune” *nonpromising nodes*
  - DFS stops exploring subtrees rooted at nodes that cannot lead to a solution and backtracks to such a node’s parent to continue the search

### 4 Example: $N$ -Queens Problem

- Place  $N$  queens on an  $N \times N$  chess board so that no two of them are in the same row, column, or diagonal



## 5 State-Space Tree of the 4-Queens Problem

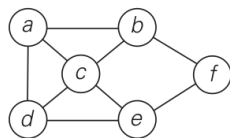


## 6 Example: Hamiltonian Circuit Problem

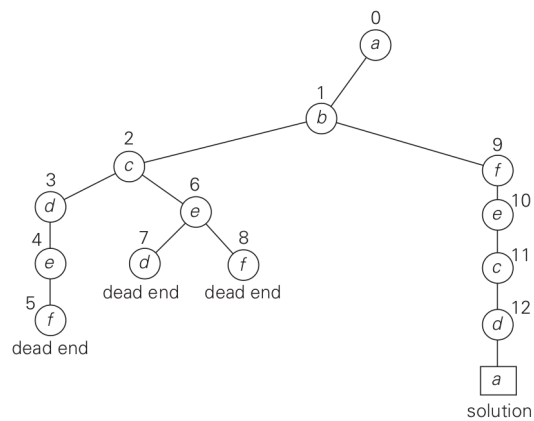
### Hamiltonian Circuit

A Hamiltonian circuit is defined as a cycle that passes through all the vertices of the graph exactly once.

- Example graph:



- State-space tree for finding a Hamiltonian circuit (numbers above the nodes of indicate the order in which the nodes are generated):

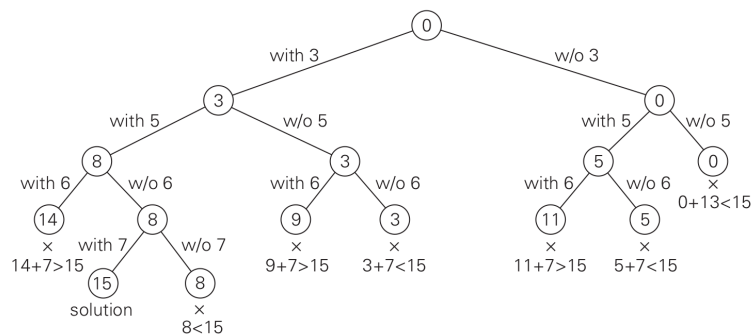


## 7 Example: Subset Sum Problem

### Subset Sum Problem

Given set  $A = \{a_1, \dots, a_n\}$  of  $n$  positive integers, find a subset whose sum is equal to a given positive integer  $d$

- State space tree for  $A = \{3, 5, 6, 7\}$  and  $d = 15$  (number in each node is the sum so far):



## 8 Algorithm

### Backtrack Algorithm

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1: procedure BACKTRACK( $X[1 \dots i]$ )
2:   ▷ Input:  $X[1 \dots i]$  specifies first  $i$  promising components of a solution
3:   ▷ Output: All the tuples representing the problem's solutions
4:   if  $X[1 \dots i]$  is a solution then
5:     write  $X[1 \dots i]$ 
6:   else
7:     for each element  $x \in S_{i+1}$  consistent with  $X[1 \dots i]$  and the
       constraints do
8:        $X[i+1] \leftarrow x$ 
9:       Backtrack ( $X[1 \dots i+1]$ )
  
```

- Output:  $n$ -tuples  $(x_1, x_2, \dots, x_n)$
- Each  $x_i \in S_i$ , some finite linearly ordered set

## 9 Think About It

- Continue the backtracking search for a solution to the four-queens problem, to find the second solution to the problem
- Explain how the board's symmetry can be used to find the second solution to the four-queens problem